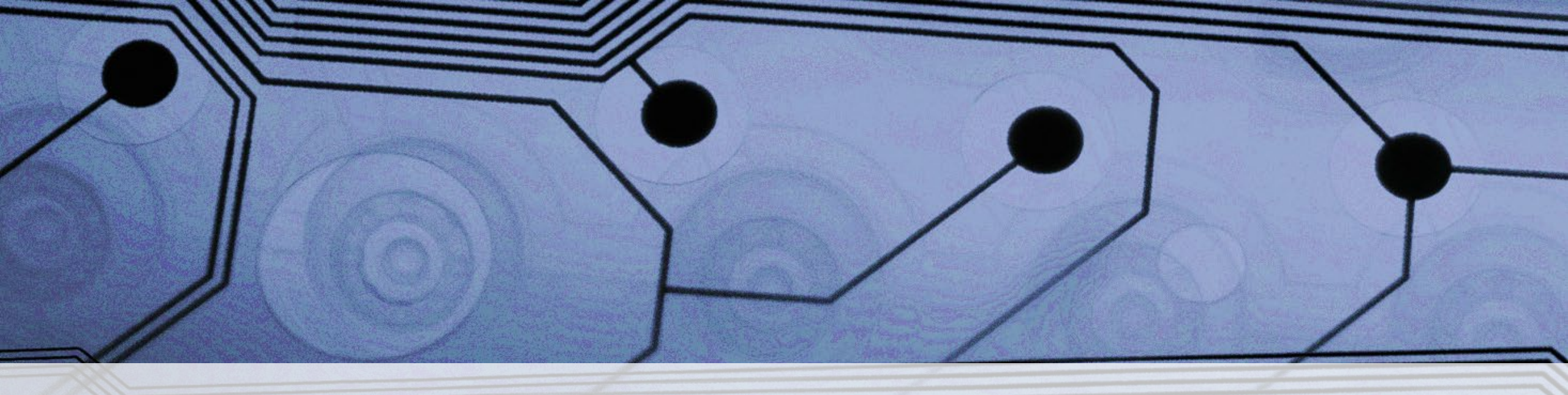
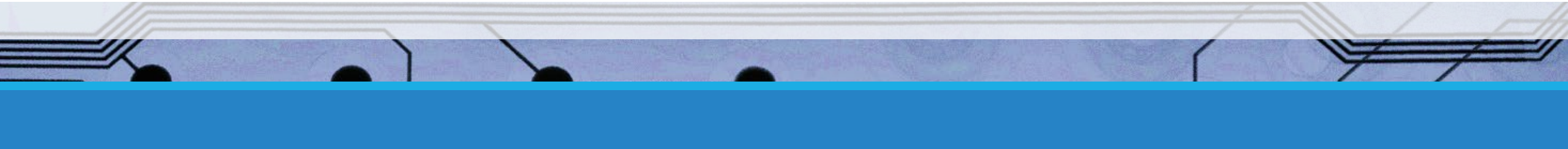




Hands-on PCB Design and Electronic Prototyping with Autodesk Fusion

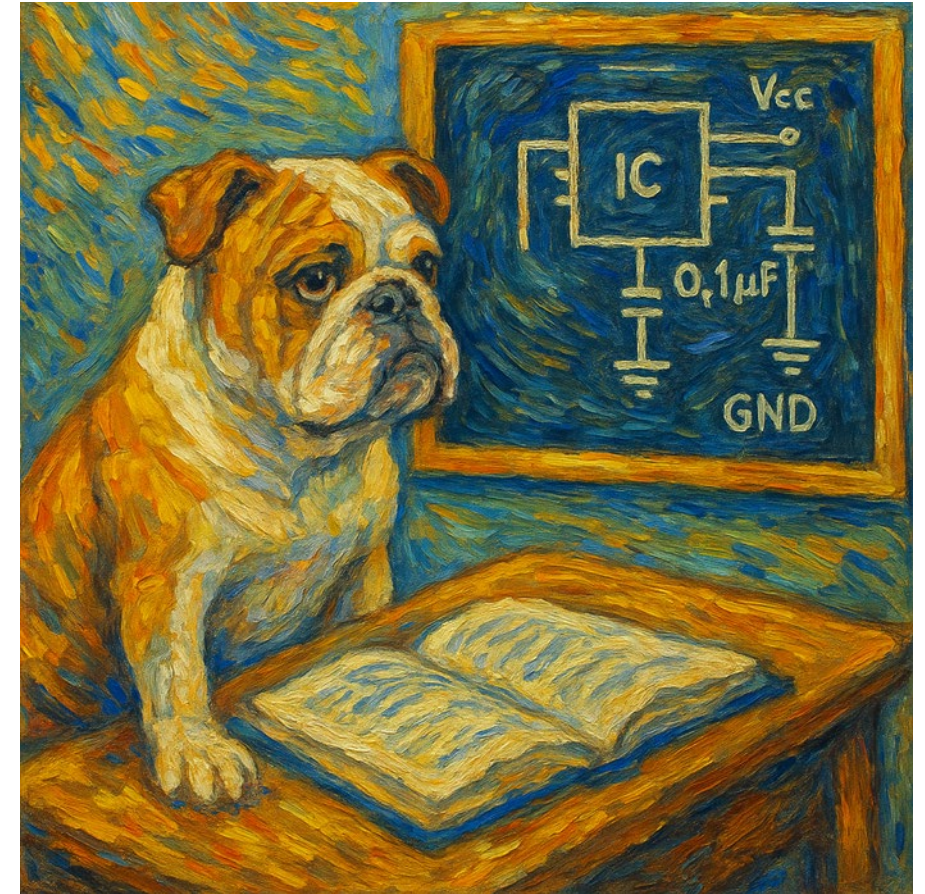
SUMMER ENRICHMENT- LYLES COLLEGE OF ENGINEERING

REVISION 1.0



Day 5: Review

- PCB Layout Essentials (Part 2)
 - Routing Basics
 - Trace routing (width, spacing, clearance rules)
 - Via usage and types
 - Design Considerations
 - Ground and power planes, copper pours, and polygon regions
 - Hands-on Activity
 - SMT Voltage Regulator Schematic/PCB



Day 6: Advanced PCB Design and Best Practices

- Signal Integrity and EMC
 - Managing crosstalk and noise
 - Proper grounding and decoupling capacitors
- High-speed and RF considerations
- Hands-on Activity
 - Use AI to recommend possible simple PCB projects
 - Select one and start designing





AI Minute



-
- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc...
 - **What is crosstalk and how does it affect signal quality in PCB circuits?**

Techniques to Reduce Crosstalk

- 1. Increase Spacing Between Traces
 - Most effective and simple fix.
 - Doubling the distance between signal traces can significantly reduce capacitive and inductive coupling.

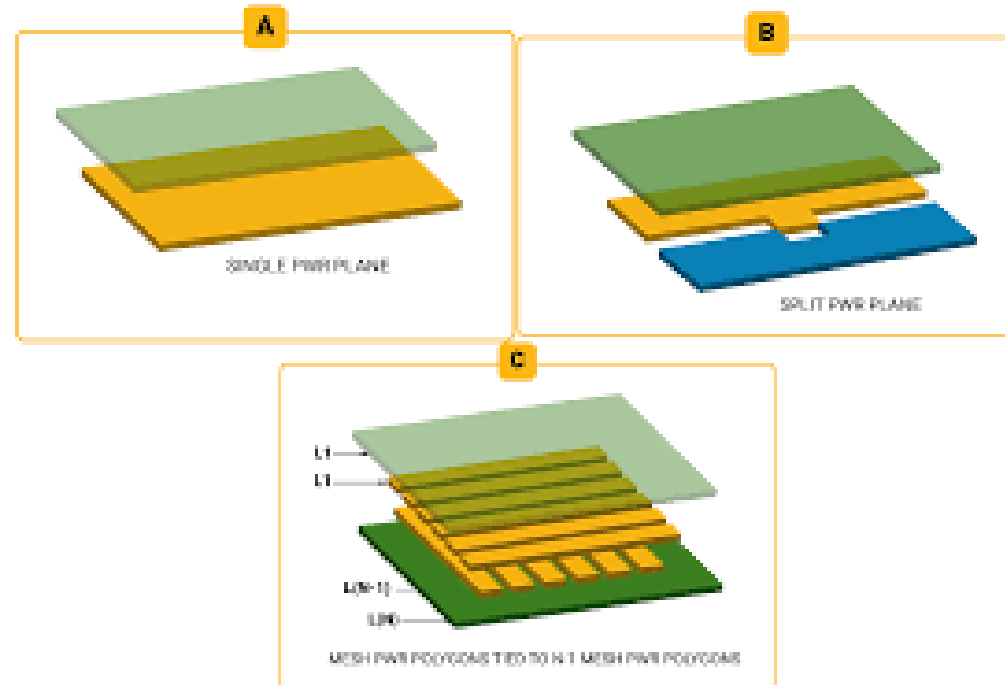
Increase Spacing
Between Traces



Techniques to Reduce Crosstalk

- 2. Use Ground Planes

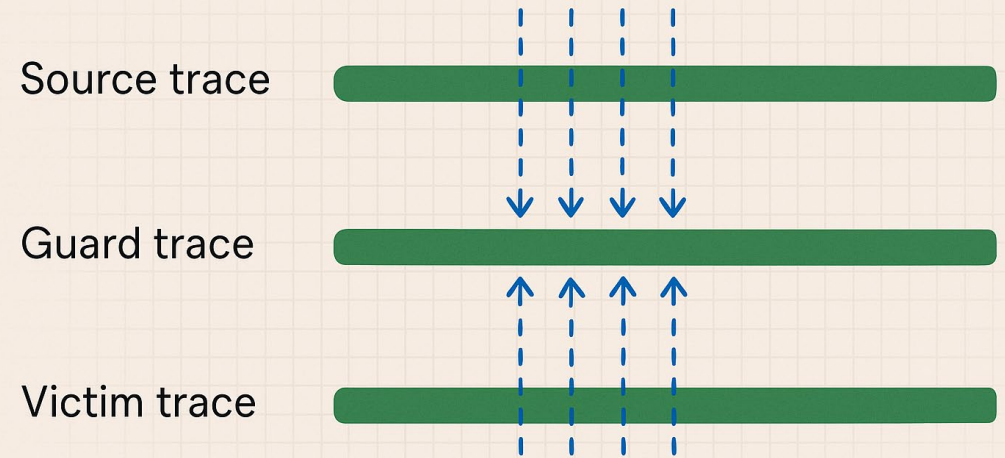
- A solid ground plane underneath signal layers provides a low-impedance return path and shields signals.
- Helps confine electric fields, reducing both capacitive coupling and loop areas for inductive coupling.



Techniques to Reduce Crosstalk

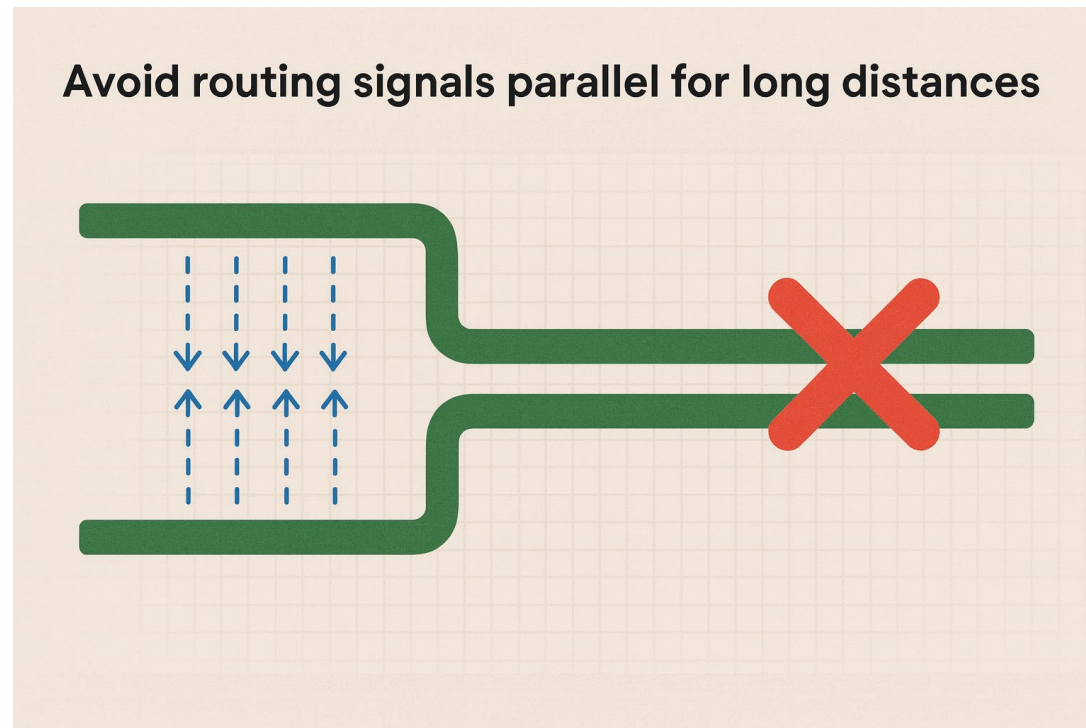
- 3. Use Guard Traces
 - Place a grounded guard trace between two high-speed signals.
 - Connect it to ground via vias at regular intervals.

Use guard traces to minimize PCB crosstalk



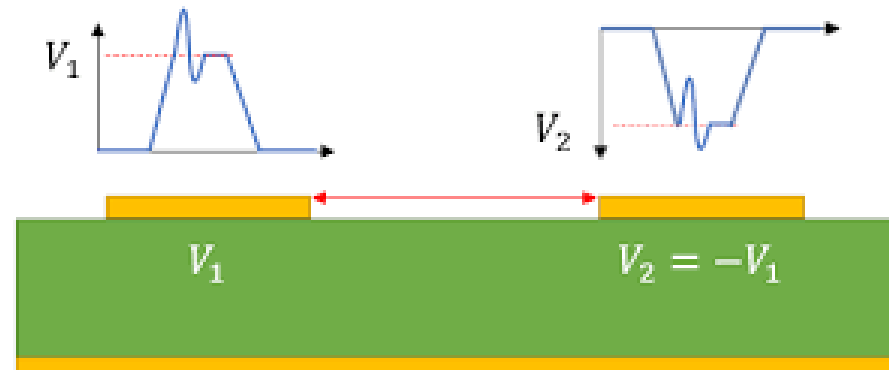
Techniques to Reduce Crosstalk

- 4. Reduce Parallelism of Signal Traces
 - Avoid routing signals parallel for long distances, especially if they switch fast or carry high-frequency signals.
 - Use short, perpendicular crossing if signals must cross each other on adjacent layers.



Techniques to Reduce Crosstalk

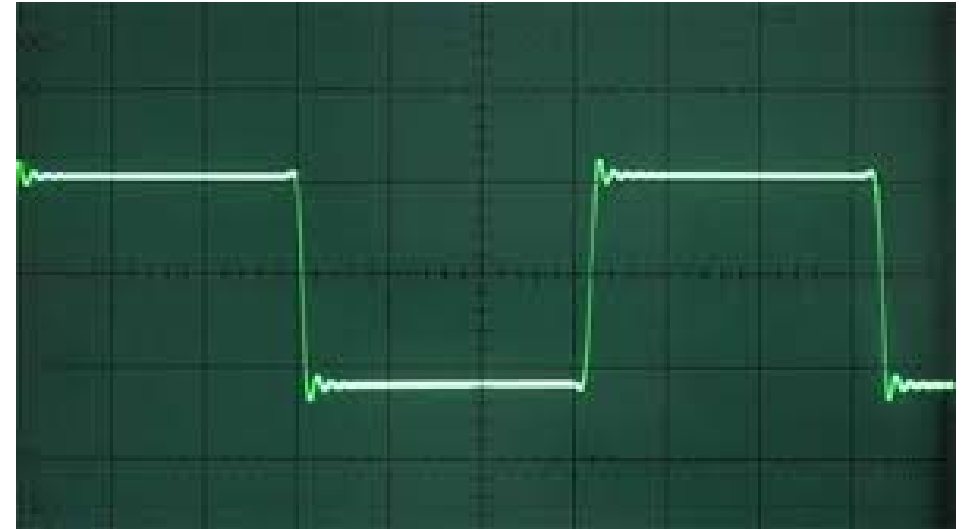
- 5. Use Differential Pairs with Controlled Impedance
 - For high-speed signals, use differential signaling (e.g., USB, HDMI, LVDS). Route pairs closely together with matched lengths and avoid unbalanced environments.
 - Controlled impedance can minimize reflections.



$$\text{Received signal: } V = (V_1 + V_N) - (V_2 + V_N) = 2V_1$$

Techniques to Reduce Crosstalk

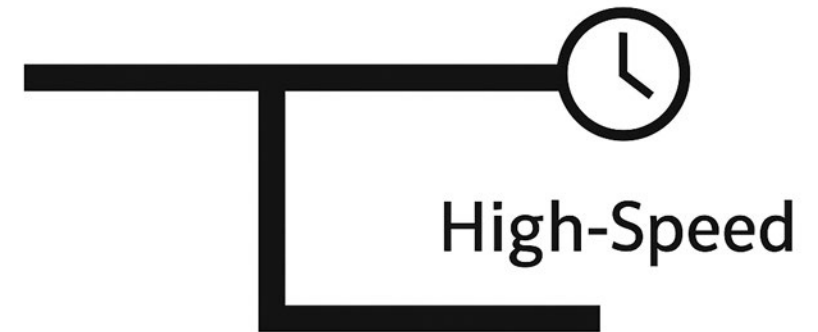
- 6. Reduce Signal Rise/Fall Times
 - Use slew rate control if your device or driver allows it.
 - Slower transitions reduce high-frequency content, which reduces coupling.



Techniques to Reduce Crosstalk

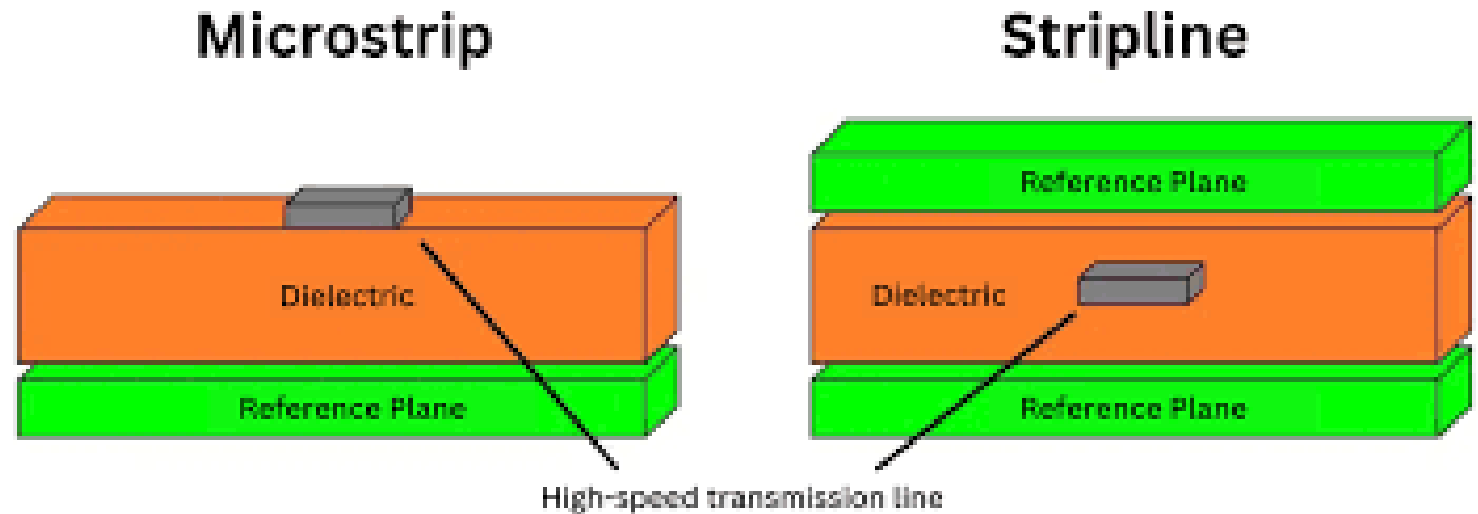
- 7. Route Clock and High-Speed Signals First
 - Give them priority in placement and isolation.
 - Keep them away from noisy or sensitive traces (e.g., analog lines).

**Route Clock and
High-Speed Signals
First**



Techniques to Reduce Crosstalk

- 8. Use Stripline Instead of Microstrip (If Possible)
 - Stripline (signal between two planes) contains fields better than microstrip (signal over a single plane).
 - Ideal for high-speed or sensitive signal routing.



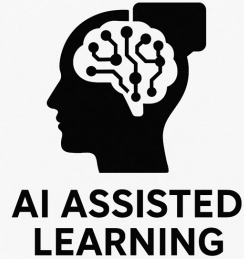
Techniques to Reduce Crosstalk

- 9. Layer Stackup Design
 - Use signal layers adjacent to planes.
 - Alternate signal and plane layers to shield and confine signals.
- Quick Example:
 - If you have two adjacent traces that carry high-speed signals:
 - Increase the spacing to at least 3x the trace width.
 - Add a grounded guard trace between them.
 - Ensure both are routed over a continuous ground plane.
 - Avoid sharp corners and maintain consistent impedance.





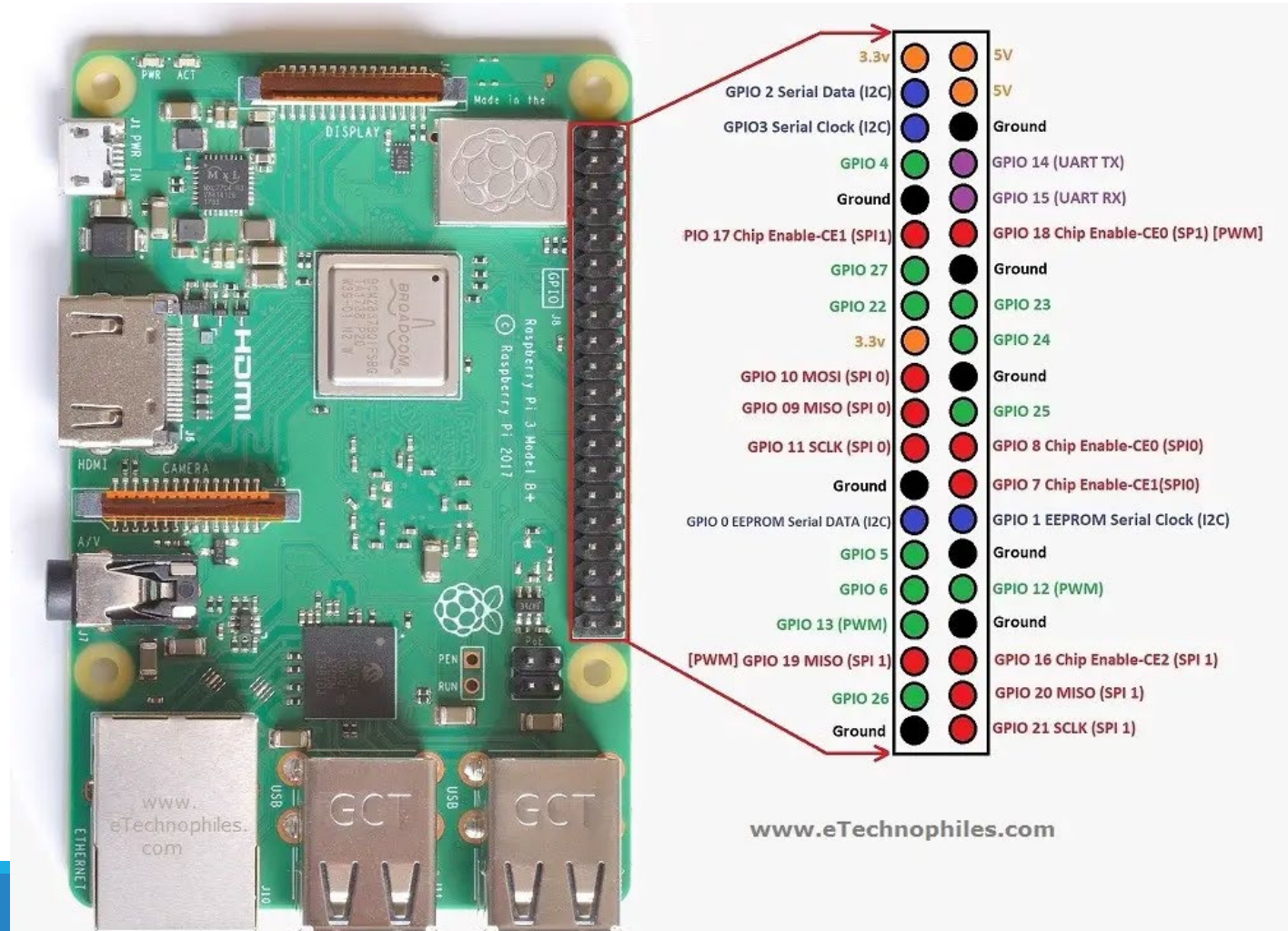
AI Minute

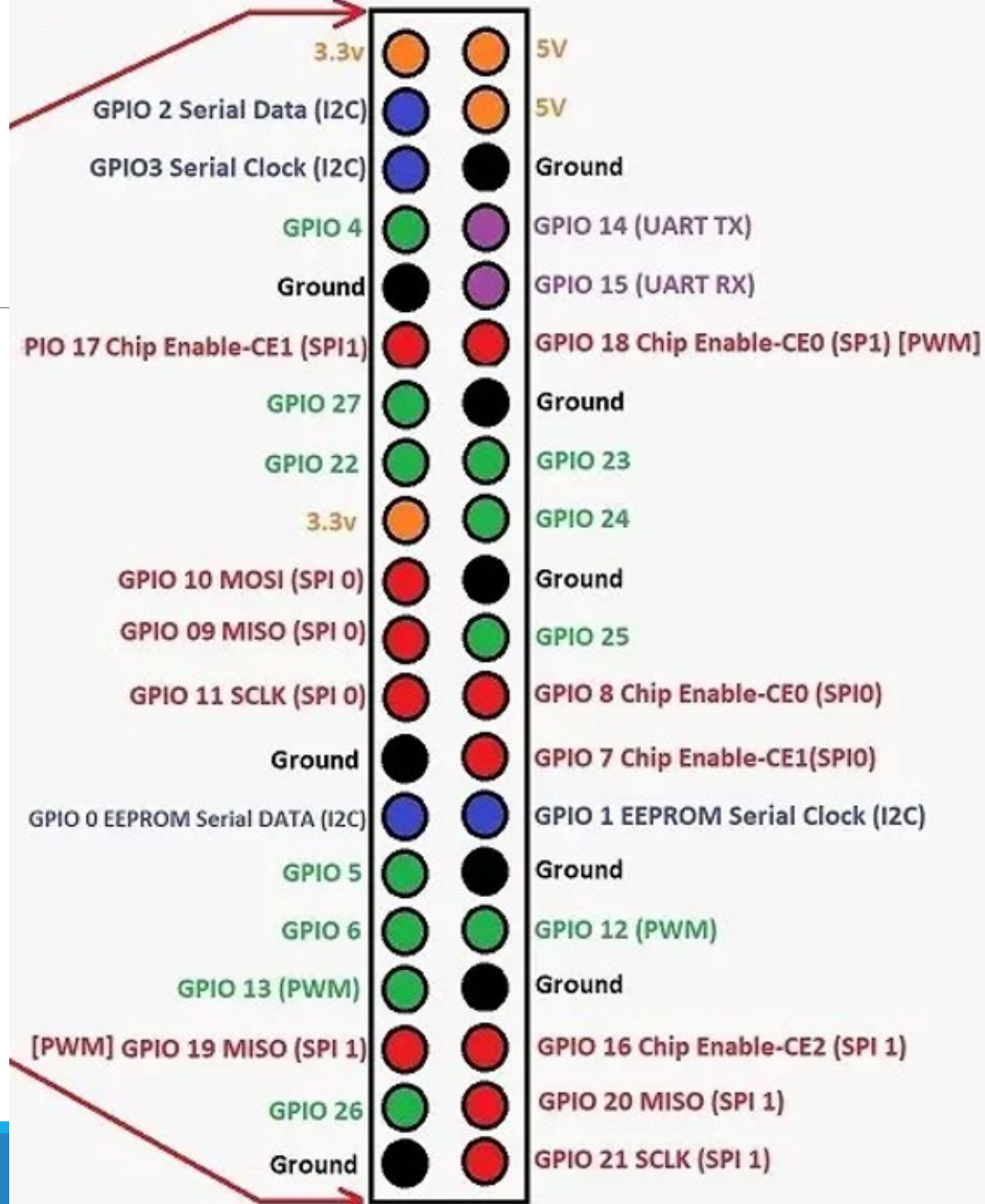


-
- Please type the following prompts into the AI Tool of your choice, Chat GPT, Gemini, Grok, Copilot, etc...
 - Please create a list of 100 project ideas for beginning university-level electrical engineering students to design a very simple, but useful, PCB?
 - Select a project on the list, or choose another project, and type something to the effect of- “Can you show a circuit for (selected project), list what parts are needed and describe how it works?”

Raspberry Pi Pinout- Create Library Component- PCB Footprint

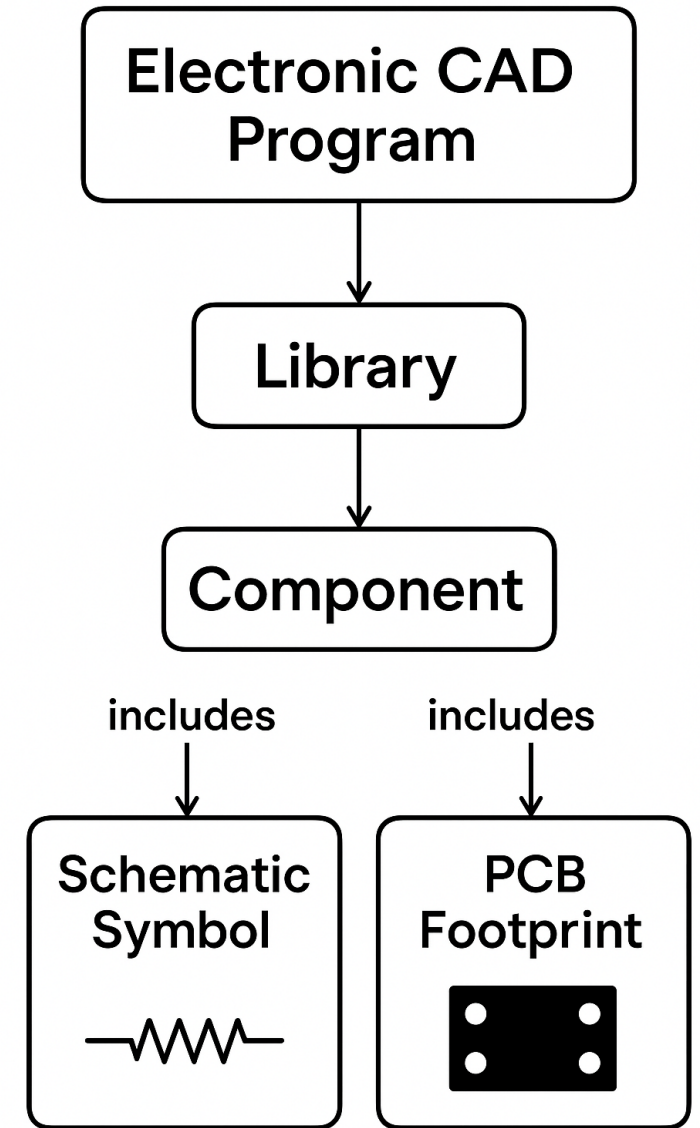
- Component Creation
 - Create Schematic Symbol
 - Create PCB Footprint
 - Combine Footprint and Symbol into a Component
- Create a new project
- Place the newly created Raspberry Pi I/O connector component on the schematic and run ERC
- Switch to PCB and place the component on the PCB
- Adjust the PCB outline to be a more appropriate size for the connector





CAD Library Elements

- Libraries
 - Libraries contain components
- Components
 - Components include:
 - A Schematic Symbol
 - A PCB Footprint
 - A 3D Model



Questions?

